Sheth L.U.J and Sir M.V College
Python programming
Practical no:9

Aim: To study about the sets in the python programming.

1) Creating Sets in python:

```
thisset = {"Tasmiya","Summaiya","Sammrin","Sheefa" , "Guddi"}
print(thisset)
```

```
{'Tasmiya', 'Summaiya', 'Sammrin', 'Sheefa', 'Guddi'}
```

```
thisset = {"Tasmiya","Summaiya","Sammrin","Sheefa","Guddi"}
print(len(thisset))
```

```
5
```

2) Access Set items:

```
thisset = {"Keyboard","Mouse","Monitor","Speaker"}

for x in thisset:
  print(x)
```

```
Mouse
Keyboard
Speaker
Monitor
```

```
thisset = {"Keyboard","Mouse","Monitor","Speaker"}

print("banana" in thisset)
```

```
False
```

3) Add set items:

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```
thisset = {"venus","saturn","mars","earth"}

thisset.add("orange")

print(thisset)
```

```
{'orange', 'mars', 'earth', 'saturn', 'venus'}
```

```
thisset = {"apple", "banana", "cherry"}
mylist = ["rose","lotus","tulip"]

thisset.update(mylist)

print(thisset)
```

```
{'cherry', 'apple', 'lotus', 'banana', 'tulip', 'rose'}
```

4) Remove set items:

```
thisset = {"light","fan","wall","roof"}

x = thisset.pop()

print(x)

print(thisset)
```

```
fan
{'wall', 'roof', 'light'}
```

```
thisset = {"fan","light","wall","roof"}

thisset.remove("wall")

print(thisset)
```

```
{'fan', 'roof', 'light'}
```

5) Loop sets:

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```
thisset = {"cat","dog","mouse","monkey"}

for x in thisset:
  print(x)
```

```
monkey
cat
mouse
dog
```

6) Join Sets:

```
set1 = {"ab","cv","rt","ty"}
set2 = {6,7,8,9}

set3 = set1.union(set2)
print(set3)
```

```
{'ab', 6, 7, 8, 9, 'cv', 'rt', 'ty'}
```

```
set1 = {"apple", "banana", "cherry"}
set2 = {"google", "microsoft", "apple"}

set1.intersection_update(set2)

print(set1)
```

```
{'apple'}
```

```
set1 = {"apple", "banana", "cherry"}
set2 = {"google", "microsoft", "apple"}

set3 = set1.intersection(set2)
print(set3)
```

```
{'apple'}
```

7) Frozen Sets:

```
x = frozenset({"lily","levender","marigold"})
print(x)
print(type(x))
```

```
frozenset({'lily', 'marigold', 'levender'})
<class 'frozenset'>
```

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```
a = frozenset({1, 2})
b = frozenset({3, 4})
c = frozenset({2, 3})
print(a.isdisjoint(b))
print(a.isdisjoint(c))
```

```
True
False
```

```
a = frozenset({1, 2})
b = frozenset({1, 2, 3})
print(a.issubset(b))
print(a <= b)
print(a < b)
```

```
True
True
True
```

```
a = frozenset({1, 2, 3})
b = frozenset({1, 2})
print(a.issuperset(b))
print(a >= b)
print(a > b)
```

```
True
True
True
```

```
a = frozenset({1, 2, 3})
b = frozenset({3, 4, 5})
print(a.symmetric_difference(b))
print(a ^ b)
```

```
frozenset({1, 2, 4, 5})
frozenset({1, 2, 4, 5})
```

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